

Propagation of error

Sections 3.1 and 3.2

Parts of Error

We can think of the error of the measurement as being composed of two parts:

- **Systematic error (bias) - accuracy**

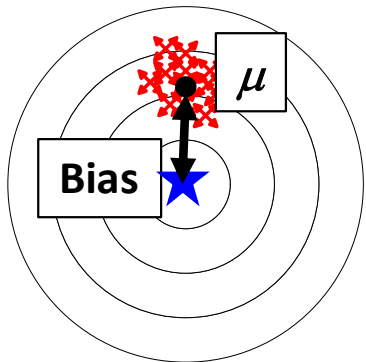
Bias is the part of the error that is the same for every measurement. For example, a scale that always gives you a reading that is too low.

- **Random error - precision**

Random error is error that varies from measurement to measurement and averages out to zero in the long run.

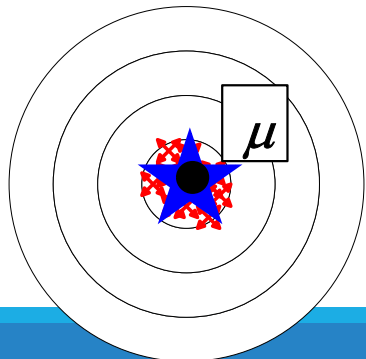
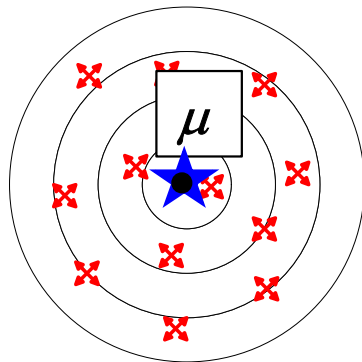
$$\text{Measured value} = \text{true value} + \text{bias} + \text{random error}$$

Propagation of error



Low uncertainty
Non-zero bias

High uncertainty
No bias



Low uncertainty
No bias

Target

→ True value ★

Shots

→ Measurements ✕

Average of shots → Mean measurement (μ) •

Spread of shots → Uncertainty (σ)

Bias

sistematik hata

yanlılık

eğilim

önyargı

taraf tutma

eğimli

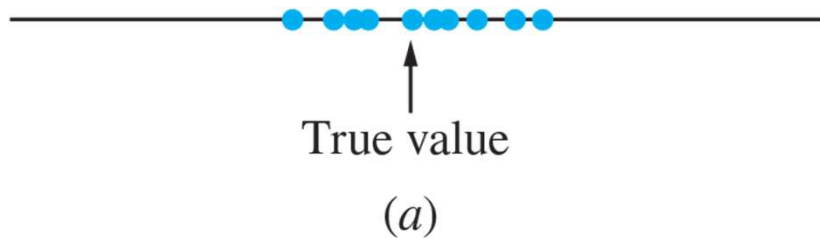
çapraz

meyil

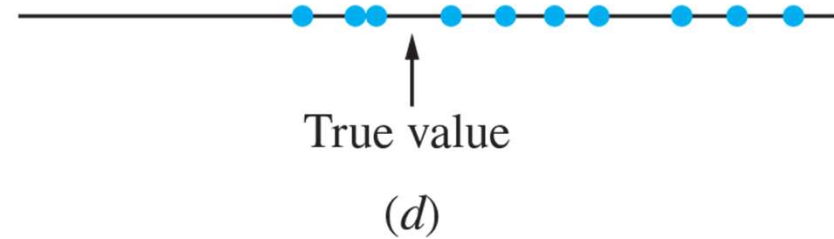
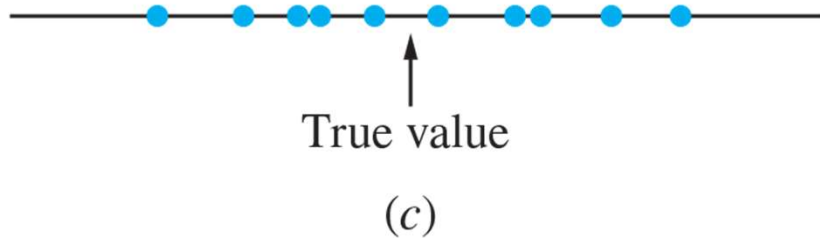
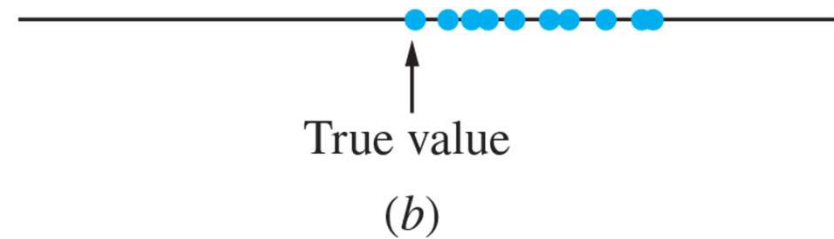
- **unbiased estimator:**
yansız kestirici
- **biased estimator:**
yanlı kestirici

In terms of samples

**Low uncertainty
No bias**



**Low uncertainty
Non-zero bias**



**High uncertainty
No bias**

**High uncertainty
Non-zero bias**

Error

A measured value is a random variable with mean μ and standard deviation σ .

The bias in the measuring process is the difference between the mean measurement and the true value:

$$\text{Bias} = \mu - \text{true value}$$

The smaller the bias, the more accurate the measuring process.

The smaller the uncertainty, the more precise the measuring process.

More on Error

Uncertainty can be estimated from repeated measurements, but in order to estimate the bias, we must have additional information about the true value.



(a)



(b)



(c)



(d)

More on Error

Let $X_1, \dots, X_n$ be independent measurements, all made by the same process on the same quantity. The sample standard deviation σ can be used to estimate the uncertainty.

Estimates of uncertainty are often crude, especially when based on small samples.

If the true value is known, the sample mean can be used to estimate the bias:

$$\text{Bias} \approx \bar{X} - \text{true value}$$

If the true value is unknown, the bias cannot be estimated from repeated measurements.

Example

A laboratory sample of gas is known to have a carbon monoxide (CO) concentration of 50 parts per million (ppm).

A spectrophotometer is used to take five independent measurements of this concentration. The five measurements, in ppm, are 51, 47, 53, 53, and 48.

Estimate the bias and the uncertainty in a spectrophotometer measurement.

Sample mean = 50.4

Sample standard deviation: 2.8